
Convergence of 1-D CHF

A clean template for a rewritten manuscript

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1 Introduction

1.1 Motivation

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Explain the one-dimensional CHF problem, its origin, and why the convergence statement is the central question of the manuscript.

1.2 Main question

Draft note

State the approximation, the candidate limit, the topology of convergence, and the parameter regime in one place.

1.3 Relation to previous work

Draft note

Summarize the input from the renormalization-flow notes and isolate the parts that will be reproved here.

1.4 Organization

Draft note

Add a short roadmap after the final section structure is fixed.

2 Setup and Notation

2.1 Probability space and noise

Draft note

Define the probability space, filtration, driving noise, and all covariance normalizations.

2.2 The approximating equation

Draft note

Write the regularized or renormalized equation, its mild formulation, and the initial condition.

2.3 The limiting object

Draft note

Define the proposed limiting process and the notion of solution used for it.

2.4 Function spaces and topology

Draft note

Record the spatial domain, norms, path spaces, and the topology in which convergence will be proved.

2.5 Scaling and parameters

Draft note

Collect every small parameter, rescaling, stopping time, and convention in a single table or definition.

3 Main Results

3.1 Standing assumptions

Assumption 3.1 (Coefficient and initial-data assumptions). Insert the precise regularity, growth, nondegeneracy, and integrability conditions used throughout the paper.

3.2 Convergence theorem

Theorem 3.2 (Main convergence theorem). *State the full convergence result, including the mode of convergence, the time interval, the spatial topology, and any quantitative rate.*

3.3 Consequences

Corollary 3.3 (Finite-dimensional convergence). *State the immediate finite-dimensional consequence of the main theorem.*

Remark 3.4. List the parts of the theorem that are genuinely one-dimensional and the parts expected to extend to other settings.

4 Preliminary Estimates

4.1 Heat-kernel estimates

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Collect the deterministic convolution, smoothing, and gradient estimates.

4.2 Stochastic integral estimates

Draft note

State the isometry, Burkholder–Davis–Gundy estimates, and moment bounds in the exact form needed later.

4.3 Well-posedness and uniform moments

Proposition 4.1 (Uniform moment bound). *Insert the estimate that is uniform in the approximation parameter.*

4.4 Renormalization-flow input

Draft note

State each result imported from the existing renormalization-flow project and record its hypotheses explicitly.

5 Proof of Convergence

5.1 Approximation scheme

Draft note

Introduce the intermediate processes and display the complete approximation chain before estimating individual errors.

5.2 Tightness or compactness

Proposition 5.1 (Tightness). *State the uniform spatial and temporal increment estimates and the resulting tightness conclusion.*

5.3 Identification of subsequential limits

Lemma 5.2 (Identification). *Specify the martingale problem, weak formulation, or limiting integral equation satisfied by every subsequential limit.*

5.4 Uniqueness of the limit

Draft note

Give the uniqueness argument that upgrades subsequential convergence to convergence of the full family.

5.5 Quantitative error bounds

Draft note

Track the separate approximation errors and optimize the intermediate scale to obtain the final rate.

5.6 Proof of the main theorem

Proof of the main convergence theorem. Assemble the preceding propositions and lemmas here. $\square$

6 Extensions and Open Questions

6.1 Alternative initial data

6.2 Other nonlinearities

6.3 Higher-dimensional comparison

6.4 Open problems

Draft note

Separate conjectural extensions from statements that follow directly from the proof.

A Auxiliary Results

A.1 Analytic estimates

Draft note

Move long heat-kernel calculations and deterministic inequalities here.

A.2 Probability tools

Draft note

Record any continuity theorem, stochastic integration result, or coupling lemma used in the main text.

A.3 Notation index

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Add a notation table once the manuscript stabilizes.